Preparing Educators for AI-Infused Classrooms

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SITE 2026

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Meet the panel

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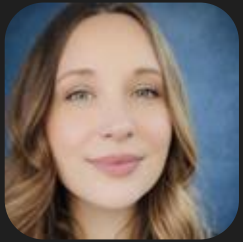

Tiffany Clemins
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Becky Osterfeld
Beavercreek City Schools, OH

Steve Geddis
Brockton Public Schools, MA

Linda Rich
We Lead CS, KY

MARSHALL

Who's Who in the Room?

Please join our Mentimeter at this time to tell us a little bit more about who you are before we begin.

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Now please share with us some of the words and phrases that come to mind when you think about Generative Artificial Intelligence (Gen AI) in education.

System-Level Gaps in AI Preparation

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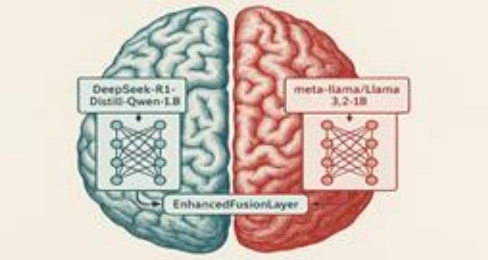
- Growing disconnect between AI-rich classrooms and preservice preparation
- AI is reshaping writing, assessment, and academic integrity
- Increased demand for metacognition and cognitive engagement
- AI must be positioned as a thinking scaffold
- Risk of widening gaps without intentional, equity-centered design
- Preservice teachers need experience with:
  - Instructional design using AI
  - Feedback and revision with AI
  - Equity and access considerations
  - Comparative reasoning and critical thinking tasks
- Teacher prep programs must embed AI into authentic pedagogical practice



Policy and Equity

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- Policy factors:
    - Infrastructure shapes opportunity.

    - Teacher readiness matters.

    - Support must be sustained.

    - Local context should guide AI use.

Classroom Realities & Instructional Coaching

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- Never enough time
- Need for educator Gen AI literacy is great
- Malcolm Knowles and andragogy
  - Involved learners
  - Problem-centered learning
  - Realistic learner experiences
  - Relevance to learners' educational lives

AI, Accessibility, and Special Education

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- AI is already in classrooms, but many teachers—especially in special education—are not fully trained or supported in using it effectives
- Special educators face heavy workloads (IEPs, data tracking, differentiation), and AI can help streamline these tasks
- Start educating/training pre-service teachers in AI to make them more efficient with the paperwork tasks to free up time for what matters most: **individualized instruction, relationships, and student self-advocacy**
- AI can support differentiation, accessibility, and progress monitoring, making learning more equitable for students with disabilities
- However, AI cannot replace teacher expertise and must always align with **IEP goals, ethical use, and student needs**
- To be effective, AI use requires **ongoing training, equitable access, and intentional integration into real classroom practice**

Technical Fluency, STEM/CTE, and Cross-Disciplinary Literacy

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- Generative AI cannot be ignored in future workforce preparation
- Educators can teach ethical AI use while integrating technical tools
- AI supports learning, scaffolding, and accessibility (e.g., ELL support)
- AI literacy is important across CTE fields, not just computer science
- Goal: future-ready classrooms that foster innovation and critical thinking

You are welcome to continue submitting questions as we talk!

Discussion

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Teacher Use of Metacognitive Practices in AP Classrooms

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National Social Science Association
Fall Conference
Hot Springs, AR
Virtual Presentation
October 19-21, 2025

Presentation Goal

Develop an initial foundation for a larger study of teacher practices in the use of metacognitive strategies and learning in advanced curricula in secondary schools.

Metacognition: What is It?

Knowledge about one's own cognitive processes and the ability to monitor and control these processes. (Flavell, 1976).

- *Metacognitive Knowledge* is awareness and regulation of one's cognitive processes.
  - What strategies work best for different tasks and when to use them
  - One's own thinking limitations as a learner
- *Metacognitive Regulation* encompasses planning, monitoring, and evaluating learning.
  - Planning how to approach tasks
  - Monitoring comprehension progress
  - Evaluation the effectiveness of strategies

Flavell (1976); Steiner (2023)

History of Metacognition

- Introduced by John Flavell in the 1970s.
  - The concept of thinking about thinking can be traced back to ancient philosophers like Aristotle and St. Augustine. John Locke also discussed children reflecting on their own thinking. Interest in metacognition grew significantly in the late 20th century through the work of researchers like Ann Brown, who made the distinction between knowledge of cognition and regulation of cognition.
- Rooted in constructivist theory (Piaget, Vygotsky).
- Connected to Dewey's emphasis on reflective learning.

(Flavell, 1976; Dewey, 1910; Vygotsky

Metacognitive Learning Skills (MLS)

Metacognitive learning skills are application of metacognition-behaviors appropriate for and relevant to improving performance on a specific learning task.

- Self-regulatory strategies: Use metacognitive insights to guide learning (self-assess, monitor, select appropriate task-specific strategies)
- Task Oriented Techniques: Ways learners interact with content to help them remember, learn, think, and understand; may be content or context specific; can be taught as methods

Flavell (1976); Dewey (1910); Vygotsky (1978)

Benefits of Metacognitive Learning

- Knowledge of Cognition: Refers to what learners know about their own thinking, including:
  - Declarative knowledge - awareness of personal learning strengths, weaknesses, and preferences.
  - Procedural knowledge - understanding which strategies are effective for different tasks.
  - Conditional knowledge - knowing when and why to use specific strategies.
- Regulation of Cognition: Involves the active management of one's learning processes. Key components include:
  - Planning - setting goals, selecting strategies, and allocating resources before learning begins.
  - Monitoring - tracking comprehension and task performance during learning.
  - Evaluating - reflecting on outcomes and strategy effectiveness after learning.

Hacker, Dunlosky, & Graesser (2009)

MARSHALL

Metacognition and Advanced Secondary Curriculum

- Advanced courses demand deeper thinking
  - Students must engage in analysis, synthesis, and evaluation and not just memorization
- Metacognitive skills are often assumed
  - Many students enter AP and honors courses without explicit instruction in how to manage their own learning
- Instruction should make thinking visible
  - Teachers can support metacognitive growth by modeling strategies, thinking aloud, and prompting student reflection

Conley (2014); McGuire (2021)

Process for Developing Metacognitive Skills

- Embed planning, monitoring, and evaluation in lessons.
- Use scaffolds like reflection prompts and checklists.

Metacognitive Process	Example in Practice
Plan	Student sets learning goals; teacher models task analysis
Monitor	Student checks understanding; teacher gives formative feedback
Evaluate	Student reflects; teacher guides self-assessment
Adjust	Student revises strategy; teacher scaffolds reflection

Steiner (2023); National Research Council (2000)

Student Barriers to Metacognition

- Pausing to self-assess is not typical behavior
- Time commitment required to pause and reflect
- Lack of awareness of effective MC strategies
- Avoidance of alternate strategies b/c time required
- A focus on finishing rather than learning
- Distance between short- and long-term goals

(Steiner, nd)

Metacognitive Learning Strategies

- Reflection journals, self-questioning, and learning logs.
- Prediction and confidence-rating exercises.
- Concept mapping and annotation.
- Peer teaching and collaborative metacognition.

(Malamed, 2021; Steiner, 2023)

Metacognitive Teaching Strategies

- Model cognitive processes aloud.
- Use reflective questioning to prompt reasoning.
- Scaffold self-assessment with rubrics and exemplars.
- Embed revision and feedback cycles.

(EEF, 2019; Steiner, 2023)

Monitoring/Assessing Student Metacognitive Learning

- Annotate course text with commentary/questions
- Assign “minute” or “muddiest point” papers
- Use guided reading questions as supplements
- Have students generate potential test questions
- Teach students how to make study guides
- Use student “confidence judgements” on tests
- Compare received and predicted grades
- Student performance analysis of assignments
- Students formally respond to teacher feedback

(Steiner, nd)

Implications for Teacher Prep & PD

- Embed metacognitive pedagogy in teacher education programs.
- Provide reflective PD structures mirroring metacognitive processes.
- Encourage iterative reflection and adjustment.

(Steiner, 2023; Conley, 2014)

Metacognition and Policy Implications

- Incorporate MC skills/strategies into state standards
- Adopt definition CCR that includes MC skills
- Develop assessments that gauge MC skills/behaviors
- Focus on learning process as well as outcomes
- Focus on student development of MC skills/strategies
- Encourage innovation and experimentation
- Expand/intensify research about MC learning/skills
- Integrate MC assessments at the state/local levels
- Encourage P-12 & IHE collaboration
- Integrate MC learning into educator prep and PD

(Conley, 2014)

Continuing Challenges

- Limited teacher training in metacognitive instruction.
- Need for research in diverse, advanced academic contexts.
- AI-supported reflection and feedback tools as emerging area.

(Steiner, 2023; McGuire, 2021)

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